# Lecture 27
# Power in steps; two proportions

# Review

**1 What is the difference between a Type I and a Type II error?**

Type I: reject $H_0$ when $H_0$ is true (a false alarm, probability $\alpha$). Type II: do not reject $H_0$ when $H_0$ is false (a miss, probability $\beta$)

**2 Define power.**

the probability the test rejects $H_0$ when $H_0$ is false: power = $P(\text{reject } H_0 \mid H_0 \text{ false}) = 1 - \beta$

**3 What are two ways we can increase power?**

a larger sample n (a smaller SE); a larger $\alpha$ (at the cost of more Type I errors). Also: a smaller $\sigma$, or a bigger true effect

# Calculating power: the steps

1. set up

   **$H_0$: $\mu = \mu_0$, $H_A$, $\alpha$, $\sigma$, $n$, and one possible truth: $\mu = \mu_{true}$**

2. the SE

   **$SE = \sigma / \sqrt{n}$: the same whether $H_0$ is true or not**

3. the cutoff, from $H_0$

   **upper-tailed: $\bar{x}^* = \mu_0 + Z_{1-\alpha} \times SE$; lower-tailed: $\bar{x}^* = \mu_0 + Z_{\alpha} \times SE$**

4. switch to the truth

   **$z^* = (\bar{x}^* - \mu_{true}) / SE$: the cutoff, measured from the possible truth**

5. the power

   **upper: $P(Z \geq z^*)$; lower: $P(Z \leq z^*)$; then $\beta = 1 - power$**

# Practice: harder rebar?

Steel rebar holds the concrete in bridges and buildings together, and harder bar is stronger bar. A mill's standard bar has a mean Brinell hardness of 170 kgf/mm², normally distributed with σ = 10. The mill hopes a new heat treatment raises the mean, so an engineer will test 25 treated bars: $H_0$: μ = 170 vs $H_A$: μ > 170. Before testing a single bar: if the treatment really raises the mean to 175, how likely is the test to detect it? (adapted from Penn State STAT 415, Example 10.4)

**σ = 10; n = 25; α = 0.05; one possible truth: μ = 175**

**(a) Find the SE and the cutoff $\bar{x}^*$.**

**(b) Find $z^*$ and the power. What is β?**

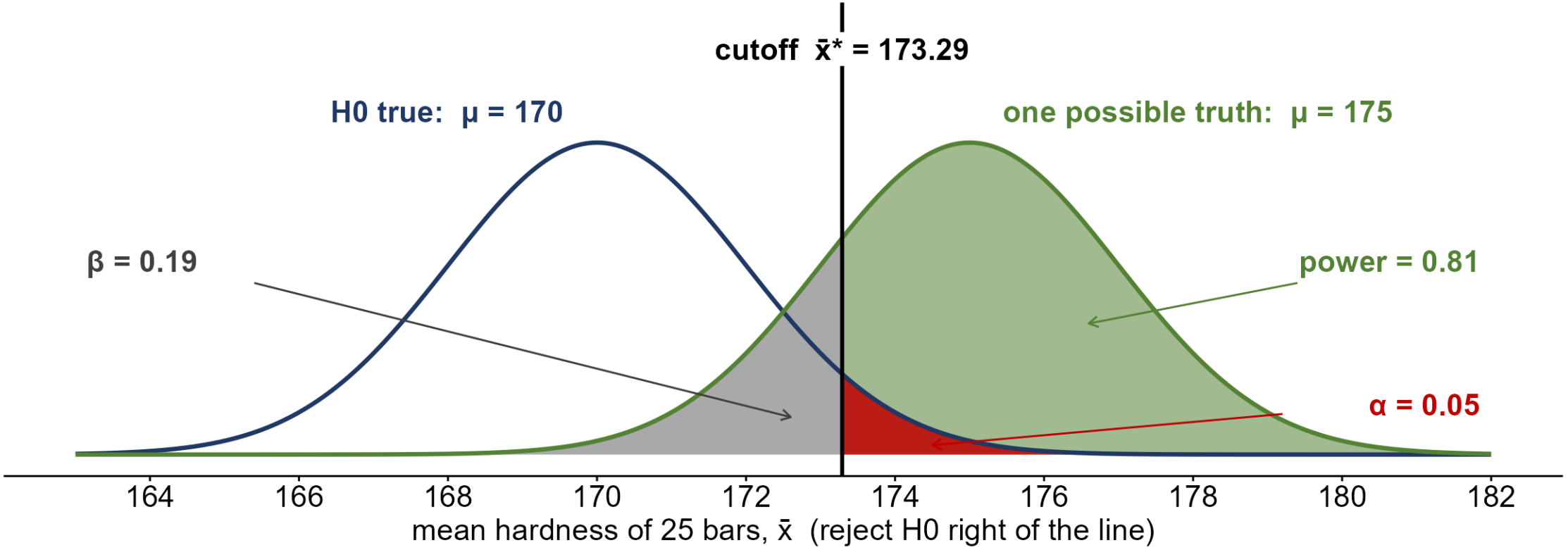
**(c) Repeat (a) and (b) with 64 bars.**

**(a) SE = 10 / √25 = 2; $\bar{x}^* = 170 + 1.645 \times 2 = 173.29$**

**(b) $z^* = (173.29 - 175) / 2 \approx -0.86$; power = $P(Z \geq -0.86) = 1 - 0.1949 = 0.81$; β = 0.19**

**(c) SE = 1.25; $\bar{x}^* = 172.06$; $z^* = -2.35$; power = 1 − 0.0094 = 0.99**

# The rebar test's power




**with 25 bars, the test catches a true mean of 175 81% of the time (power) and misses it 19% (β)**

# One population or two?

**1** **Is the mean commute of UI students more than 20 minutes?**

one sample: one population, one mean, compared with a claimed value

**2** **Do patients on a new drug recover more often than patients on a placebo?**

two samples: two populations (drug, placebo); compare $p_1$ and $p_2$

**3** **Do Moscow and Pullman voters support a new levy at the same rate?**

two samples: two populations (the two towns); compare $p_1$ and $p_2$

# Two-sample tests

what they compare

**the same parameter in two populations: $p_1$ vs $p_2$, or $\mu_1$ vs $\mu_2$**

two variables

**an explanatory variable (which group) and a response variable (the outcome)**

the parameter

**the difference: $p_1 - p_2$ or $\mu_1 - \mu_2$**

the usual null

**no difference: $H_0$: $p_1 - p_2 = 0$ (or $\mu_1 - \mu_2 = 0$)**

# Independent vs dependent samples

independent samples

**one sample tells us nothing about the other:  separate random samples, or randomized groups**

dependent samples

**each value is paired with one in the other sample:  the same subjects twice, twins**

why it matters

**pairing changes the SE, so each design gets its own test**

today

**independent samples;  dependent (paired) samples come on Wednesday**

# Practice: independent or dependent?

**1** **The blood pressure of 30 patients, before and after a new medication**

dependent: the same 30 patients measured twice (matched pairs)

**2** **Exam scores of 40 randomly chosen students from each of two sections**

independent: two separate random samples

**3** **Twin pairs: one twin gets a tutoring program, the other does not**

dependent: each child is matched with their twin

# Two proportions: the notation

populations 1 and 2

**the two groups we compare (for example, medication A and medication B)**

$p_1$, $p_2$

**the population proportions with the trait: the parameters**

$n_1$, $X_1$; $n_2$, $X_2$

**the sample sizes, and the counts of successes in each sample**

$\hat{p}_1 = X_1 / n_1$, $\hat{p}_2 = X_2 / n_2$

**the sample proportions: $\hat{p}_1 - \hat{p}_2$ estimates $p_1 - p_2$**

# Two proportions: the hypotheses

$H_0$: $p_1 = p_2$

**the same as $H_0$: $p_1 - p_2 = 0$: no difference between the populations**

$H_A$: $p_1 - p_2 > 0$

**upper-tailed: population 1 has the larger proportion**

$H_A$: $p_1 - p_2 < 0$

**lower-tailed: population 1 has the smaller proportion**

$H_A$: $p_1 - p_2 \neq 0$

**two-sided: a difference in either direction**

# The sampling distribution of $\hat{p}_1 - \hat{p}_2$

the estimate

**$\hat{p}_1 - \hat{p}_2$: the difference between the two sample proportions**

its center

**$p_1 - p_2$: on average, the estimate hits the true difference**

its SE

**$\sqrt{[\, p_1(1 - p_1) / n_1 \; + \; p_2(1 - p_2) / n_2 \,]}$: variances add for independent samples**

its shape

**approximately normal when both samples are large**

# If $H_0$ is true: pool the samples

one common p

**if $p_1 = p_2$, both samples estimate the same p, so we combine them**

the pooled proportion

**$\hat{p} = (X_1 + X_2) / (n_1 + n_2)$: all the successes over all the observations**

the SE if $H_0$ is true

**$SE_0 = \sqrt{[\ \hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)\ ]}$**

the test statistic

**$z_{obs} = (\hat{p}_1 - \hat{p}_2 - 0) / SE_0$: $z_{obs} \sim N(0, 1)$ if $H_0$ is true**

# Assumptions and the P-value

the samples

**two independent random samples, one from each population**

large enough

**$n_1\hat{p}$, $n_1(1-\hat{p})$, $n_2\hat{p}$, $n_2(1-\hat{p})$ all ≥ 15, using the pooled $\hat{p}$**

upper or lower

**$H_A$: $p_1 - p_2 > 0$: $P(Z \geq z_{obs} \mid H_0 \text{ true})$; $< 0$: $P(Z \leq z_{obs} \mid H_0 \text{ true})$**

two-sided

**$H_A$: $p_1 - p_2 \neq 0$: 2 × the smaller tail**

# Data: medication for hives

| | Still had hives | No hives | Total |
|---|---|---|---|
| **Medication A** | 20 | 180 | 200 |
| **Medication B** | 12 | 188 | 200 |

Each row is a separate random sample of adults with hives. The response is whether a patient still had hives 30 minutes after taking the medication. (OpenStax, Example 10.8)

# Example: medication for hives

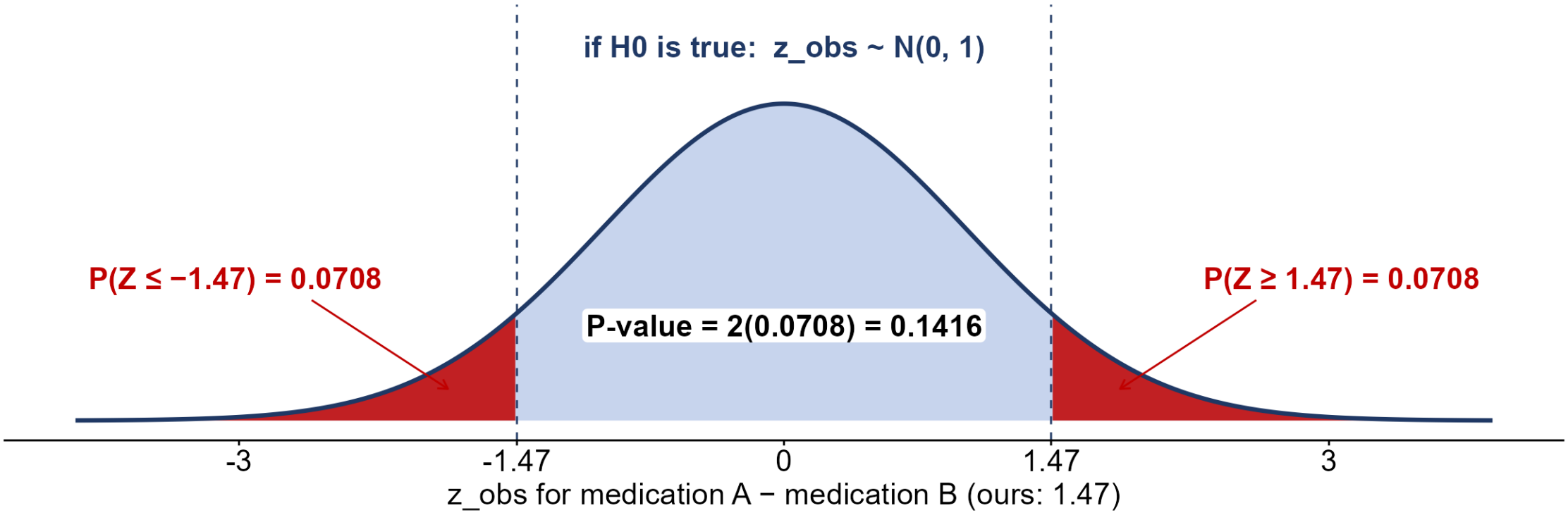
Two medications for hives are being compared. Is there a difference between them in the proportion of adults who still have hives 30 minutes after taking the medication? (OpenStax, Example 10.8)

**A: 20 of 200; B: 12 of 200; $\alpha = 0.01$**

**(a) State $H_0$ and $H_A$. Which tail?**
**(b) Find $\hat{p}_A$, $\hat{p}_B$ and the pooled $\hat{p}$. Are the samples large enough?**
**(c) Find $SE_0$ and $z_{obs}$.**
**(d) Find the P-value. What is the decision?**

**(a) $H_0: p_A - p_B = 0$ $H_A: p_A - p_B \neq 0$ (two-sided)**
**(b) 0.10, 0.06; $\hat{p} = 32 / 400 = 0.08$; smallest count $200(0.08) = 16 \geq 15$**
**(c) $SE_0 = \sqrt{[0.08(0.92)(1/200 + 1/200)]} = 0.02713$; $z_{obs} = 0.04 / 0.02713 = 1.47$**
**(d) $2(1 - 0.9292) = 0.1416 > 0.01$: do not reject $H_0$**

# The hives P-value




**0.1416 > 0.01: no evidence of a difference between the medications**

# Practice: seat belts

A researcher believes that women are less likely than men to go without a seat belt. A random sample of U.S. adults was asked whether they wear one. Is the proportion of women who do not wear seat belts lower than the proportion of men? (OpenStax, Example 10.9)

**Do not wear one: women 156 of 2,169; men 183 of 2,231; $\alpha = 0.01$**

**(a) State $H_0$ and $H_A$. Which tail?**
**(b) Find $\hat{p}_W$, $\hat{p}_M$ and the pooled $\hat{p}$. Are the samples large enough?**
**(c) Find $SE_0$ and $z_{obs}$.**
**(d) Find the P-value. What is the decision?**

**(a) $H_0: p_W - p_M = 0$ $H_A: p_W - p_M < 0$ (lower-tailed)**
**(b) 0.0719, 0.0820; $\hat{p} = 339 / 4400 = 0.0770$; smallest count $2169(0.0770) \approx 167 \geq 15$**
**(c) $SE_0 = \sqrt{[0.077(0.923)(1/2169 + 1/2231)]} = 0.0080$; $z_{obs} = -0.0101 / 0.0080 = -1.26$**
**(d) $P(Z \leq -1.26) = 0.1038 > 0.01$: do not reject $H_0$**

# Practice: electric vehicles

A marketing firm claims that younger adults are more likely than older adults to own an electric vehicle. A random sample of U.S. adults found how many in each age group own one: younger means 34 or under, older means 35 or over. (OpenStax, Example 10.10)

**Own an EV: younger 135 of 1,343; older 12 of 232; $\alpha = 0.05$**

**(a) State $H_0$ and $H_A$. Which tail?**
**(b) Find $\hat{p}_Y$, $\hat{p}_O$ and the pooled $\hat{p}$. Are the samples large enough?**
**(c) Find $SE_0$ and $z_{obs}$.**
**(d) Find the P-value. What is the decision?**

**(a) $H_0: p_Y - p_O = 0$ $H_A: p_Y - p_O > 0$ (upper-tailed)**
**(b) 0.1005, 0.0517; $\hat{p} = 147 / 1575 = 0.0933$; smallest count $232(0.0933) \approx 21.6 \geq 15$**
**(c) $SE_0 = \sqrt{[0.0933(0.9067)(1/1343 + 1/232)]} = 0.0207$; $z_{obs} = 0.0488 / 0.0207 = 2.36$**
**(d) $P(Z \geq 2.36) = 1 - 0.9909 = 0.0091 < 0.05$: reject $H_0$**